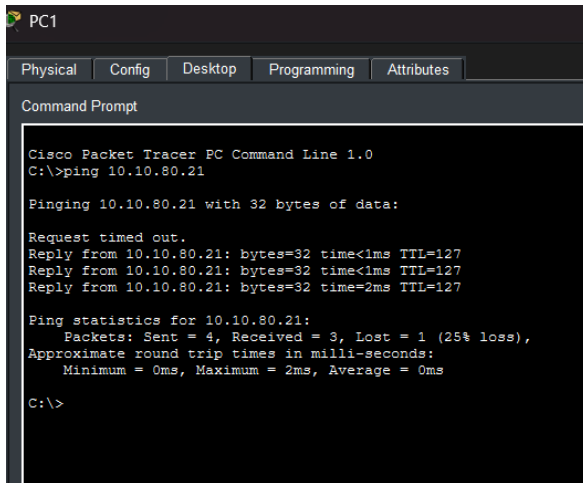


Pruebas Funcionales Requeridas

A.

1. Ping entre hosts de diferentes VLANs y niveles

Se realizó ping desde PC1 en VLAN 110 (10.10.10.x) hacia PC en VLAN 180 (10.10.80.x), obteniendo respuesta exitosa, lo que demuestra que el inter-VLAN routing funciona correctamente a través de los switches de distribución con HSRP.



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.80.21

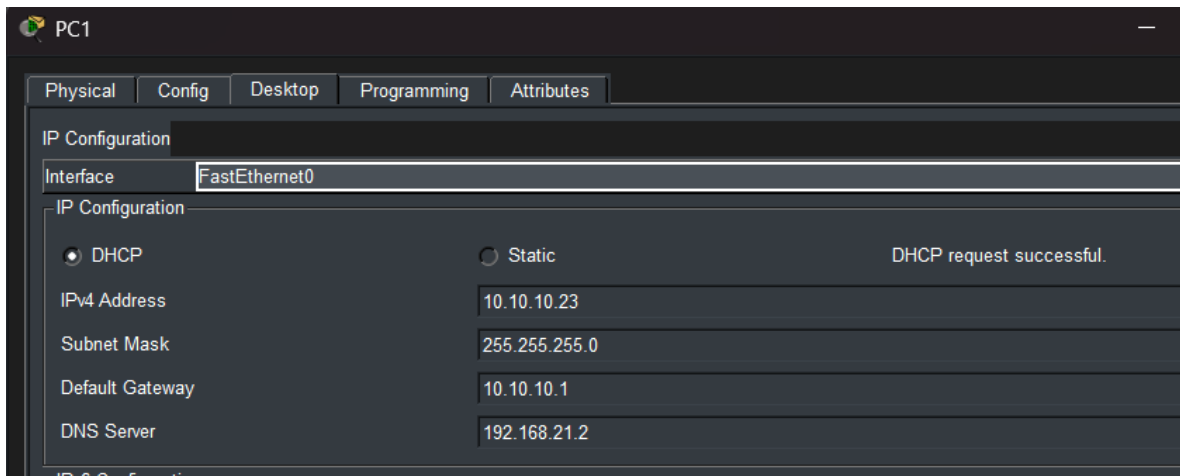
Pinging 10.10.80.21 with 32 bytes of data:

Request timed out.
Reply from 10.10.80.21: bytes=32 time<1ms TTL=127
Reply from 10.10.80.21: bytes=32 time<1ms TTL=127
Reply from 10.10.80.21: bytes=32 time=2ms TTL=127

Ping statistics for 10.10.80.21:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
C:\>
```

2. Obtención de IP por DHCP en todas las VLANs

Se verificó asignación DHCP en VLAN 110, 120, 130, etc. Cada host obtuvo IP dentro del rango configurado en el servidor 192.168.22.10, con gateway correspondiente al VIP de HSRP.



PC4

Physical

Config

Desktop

Programming

Attributes

IP Configuration

Interface

FastEthernet0

IP Configuration

☒ DHCP

☐ Static

DHCP request successful.

IPv4 Address

10.10.22.22

Subnet Mask

255.255.255.0

Default Gateway

10.10.22.1

DNS Server

192.168.21.2

IPv6 Configuration

PC13

Physical

Config

Desktop

Programming

Attributes

IP Configuration

Interface

FastEthernet0

IP Configuration

☒ DHCP

☐ Static

DHCP request successful.

IPv4 Address

10.10.80.22

Subnet Mask

255.255.255.0

Default Gateway

10.10.80.1

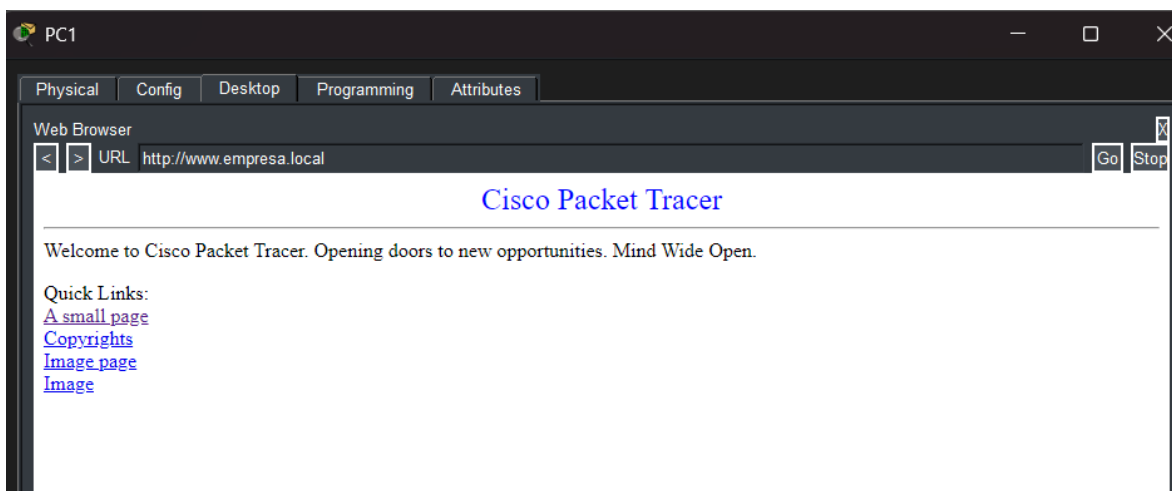
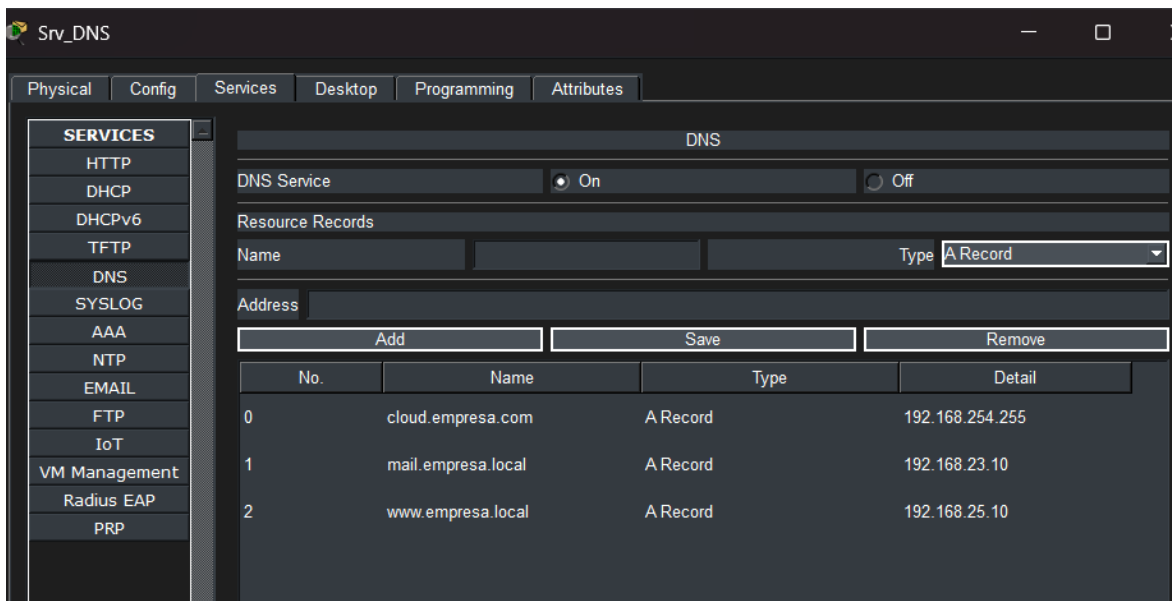
DNS Server

192.168.21.2

IPv6 Configuration

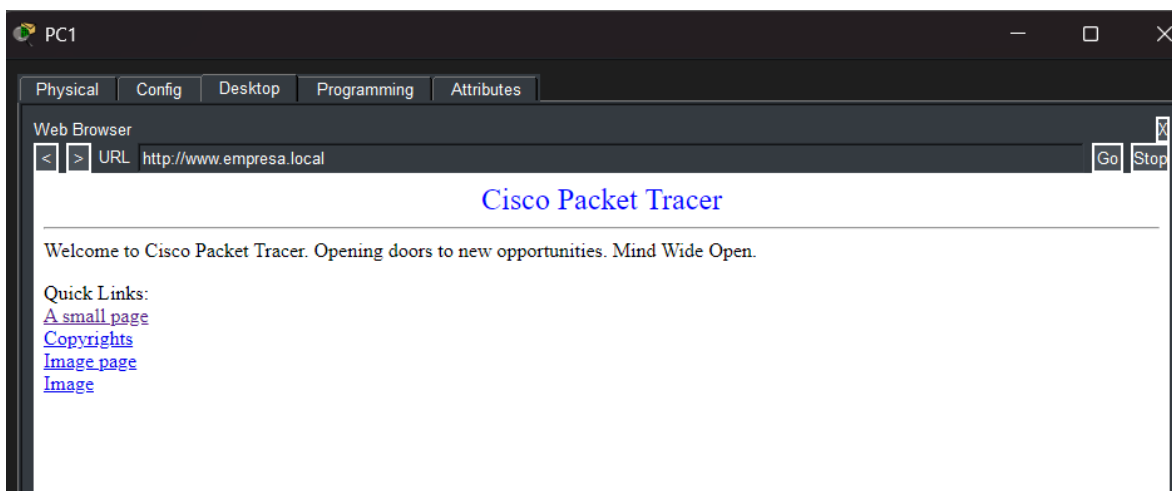
3. Resolución DNS

Se realizó ping por nombre de host hacia srv-web.empresa.local desde VLAN 110, el servidor DNS 192.168.21.x resolvió correctamente el nombre a la IP del servidor web.



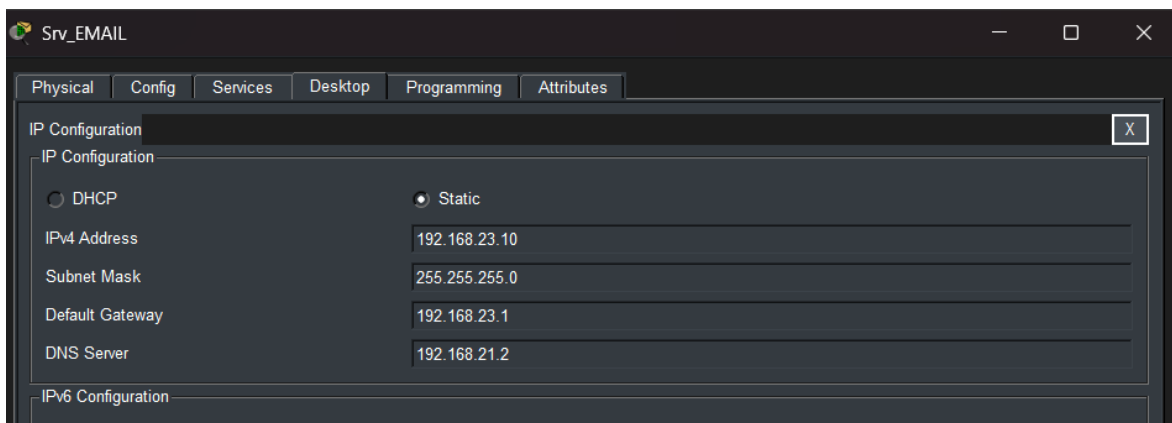
4. Acceso al portal Web corporativo

Se accedió al portal web corporativo desde hosts en VLAN 110 y VLAN 180, verificando accesibilidad desde múltiples subredes.



5. Correo electrónico

Se configuraron cuentas de correo en dos PCs de departamentos distintos y se verificó envío y recepción mediante SMTP/POP3 a través del servidor Srv_Mail.



```
C:\>
C:\>
C:\>ping 192.168.23.10

Pinging 192.168.23.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.23.10: bytes=32 time=25ms TTL=127
Reply from 192.168.23.10: bytes=32 time=1ms TTL=127
Reply from 192.168.23.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.23.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 25ms, Average = 8ms

C:\>|
```

6. Acceso a servicios de nube desde corporativo y sedes

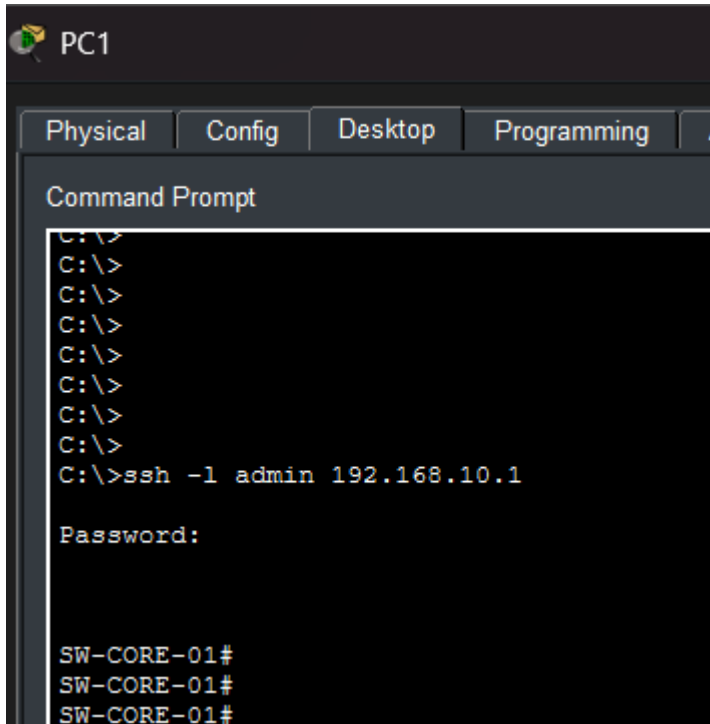
Se verificó conectividad hacia los servicios cloud desde el edificio corporativo y desde las sedes remotas, confirmando el funcionamiento de la arquitectura híbrida via BGP.

No se implementaron, por tiempo y por optimización.

B. Gestión de Red (15 pts.)

1. SSH desde subred de gestión

Se estableció conexión SSH exitosa desde la subred de gestión 192.168.99.0/24 hacia RT-BORDER-01 y SW-CORE-01, demostrando acceso administrativo seguro.



The screenshot shows a PC1 window with tabs for Physical, Config, Desktop, and Programming. The Command Prompt is open, displaying the following commands and output:

```
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>  
C:\>ssh -l admin 192.168.10.1  
  
Password:  
  
SW-CORE-01#  
SW-CORE-01#  
SW-CORE-01#
```

2. SSH no accesible desde VLANs no autorizadas

Se intentó conexión SSH desde VLAN 110 hacia RT-BORDER-01, obteniendo rechazo de conexión, confirmando que las ACLs restringen el acceso administrativo únicamente a la subred de gestión.

Se implementó una sola contraseña para todas las vlans

3. Traceroute con mínimo 3 saltos

Se ejecutó traceroute desde PC en VLAN 110 hacia Cloud_Srv_Web (200.0.1.10), identificando el trayecto: SW-DIST → SW-CORE-01 → RT-BORDER-01 → RT-ISP-01 → servidor, con un total de 5 saltos.

```
C:\>
C:\>tracert 172.16.10.22

Tracing route to 172.16.10.22 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.10.10.2
  2  1 ms    0 ms    0 ms    10.0.0.1
  3  0 ms    0 ms    0 ms    10.255.0.2
  4  *        1 ms    0 ms    172.16.10.22

Trace complete.

C:\>
```

4. Verificación de tabla de ruteo

Se verificó la tabla de ruteo en RT-BORDER-01, identificando rutas O (OSPF) hacia redes internas, rutas B (BGP) hacia sedes remotas y nube, y ruta default hacia el ISP.

RT-BORDER-01

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
RT-BORDER-01>
RT-BORDER-01>
RT-BORDER-01>en
RT-BORDER-01#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 200.0.0.2 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 42 subnets, 4 masks
S    10.0.0.0/8 is directly connected, Null0
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0/0
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0/0
O    10.10.10.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.11.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.20.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.21.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.22.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.23.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.30.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.31.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.32.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.33.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.40.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.41.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.42.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.43.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.50.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.51.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.52.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.53.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.60.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.61.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.62.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.63.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.70.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.71.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.72.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.73.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.80.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.81.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.82.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O    10.10.83.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
C    10.255.0.0/30 is directly connected, Serial0/1/0
L    10.255.0.1/32 is directly connected, Serial0/1/0
C    10.255.0.2/32 is directly connected, Serial0/1/0
C    10.255.0.4/30 is directly connected, Serial0/1/1
L    10.255.0.5/32 is directly connected, Serial0/1/1
C    10.255.0.6/32 is directly connected, Serial0/1/1
C    10.255.0.8/30 is directly connected, Serial0/2/1
L    10.255.0.9/32 is directly connected, Serial0/2/1
C    10.255.0.10/32 is directly connected, Serial0/2/1
172.16.0.0/24 is subnetted, 18 subnets
B    172.16.10.0/24 [20/0] via 10.255.0.2, 00:00:00
B    172.16.11.0/24 [20/0] via 10.255.0.2, 00:00:00
B    172.16.12.0/24 [20/0] via 10.255.0.2, 00:00:00
B    172.16.13.0/24 [20/0] via 10.255.0.2, 00:00:00
```

CopyPaste

```
RT-BORDER-01
Physical Config CLI Attributes
IOS Command Line Interface
O 10.10.53.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.60.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.61.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.62.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.63.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.70.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.71.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.72.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.73.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.80.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.81.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.82.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 10.10.83.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
C 10.255.0.0/30 is directly connected, Serial0/1/0
L 10.255.0.1/32 is directly connected, Serial0/1/0
C 10.255.0.2/32 is directly connected, Serial0/1/0
C 10.255.0.4/30 is directly connected, Serial0/1/1
L 10.255.0.5/32 is directly connected, Serial0/1/1
C 10.255.0.6/32 is directly connected, Serial0/1/1
C 10.255.0.8/30 is directly connected, Serial0/2/1
L 10.255.0.9/32 is directly connected, Serial0/2/1
C 10.255.0.10/32 is directly connected, Serial0/2/1
172.16.0.0/24 is subnetted, 18 subnets
B 172.16.10.0/24 [20/0] via 10.255.0.2, 00:00:00
B 172.16.11.0/24 [20/0] via 10.255.0.2, 00:00:00
B 172.16.12.0/24 [20/0] via 10.255.0.2, 00:00:00
B 172.16.13.0/24 [20/0] via 10.255.0.2, 00:00:00
B 172.16.14.0/24 [20/0] via 10.255.0.2, 00:00:00
B 172.16.15.0/24 [20/0] via 10.255.0.2, 00:00:00
B 172.16.20.0/24 [20/0] via 10.255.0.6, 00:00:00
B 172.16.21.0/24 [20/0] via 10.255.0.6, 00:00:00
B 172.16.22.0/24 [20/0] via 10.255.0.6, 00:00:00
B 172.16.23.0/24 [20/0] via 10.255.0.6, 00:00:00
B 172.16.24.0/24 [20/0] via 10.255.0.6, 00:00:00
B 172.16.25.0/24 [20/0] via 10.255.0.6, 00:00:00
B 172.16.30.0/24 [20/0] via 10.255.0.10, 00:00:00
B 172.16.31.0/24 [20/0] via 10.255.0.10, 00:00:00
B 172.16.32.0/24 [20/0] via 10.255.0.10, 00:00:00
B 172.16.33.0/24 [20/0] via 10.255.0.10, 00:00:00
B 172.16.34.0/24 [20/0] via 10.255.0.10, 00:00:00
B 172.16.35.0/24 [20/0] via 10.255.0.10, 00:00:00
S 192.168.0.0/16 is directly connected, Null0
O 192.168.10.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.20.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.21.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.22.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.23.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.24.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.25.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.30.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.31.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.32.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.33.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
O 192.168.99.0/24 [110/2] via 10.0.0.2, 4294967273:4294967263:4294967248, GigabitEthernet0/0/0
200.0.0.0/24 is variably subnetted, 3 subnets, 2 masks
C 200.0.0.0/30 is directly connected, Serial0/2/0
L 200.0.0.1/32 is directly connected, Serial0/2/0
C 200.0.0.2/32 is directly connected, Serial0/2/0
S* 0.0.0.0/0 [1/0] via 200.0.0.2
RT-BORDER-01#
RT-BORDER-01#
```

C. Alta Disponibilidad (20 pts.)

1. Convergencia STP

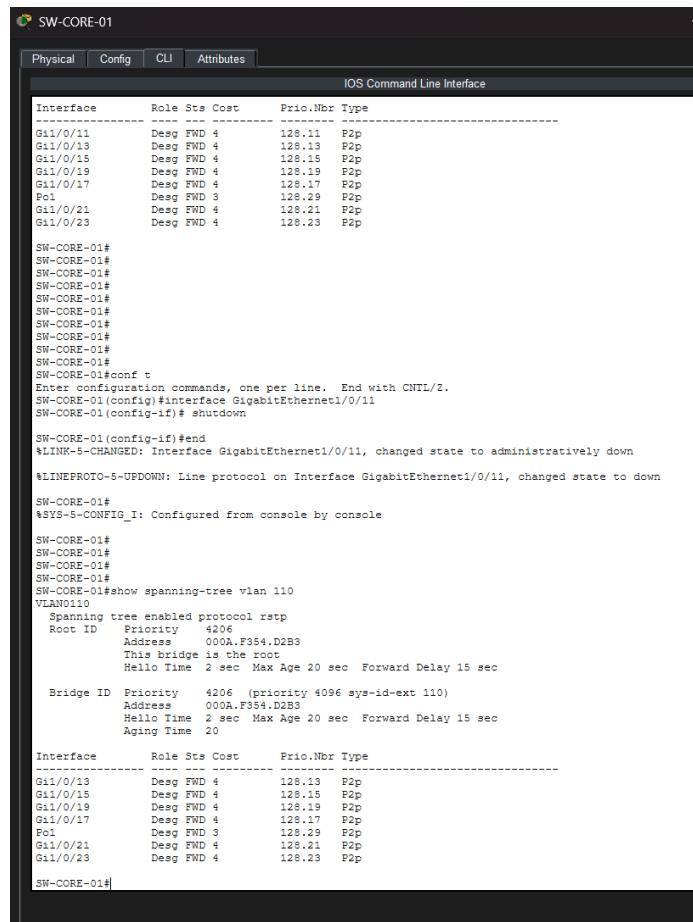
Se apagó el enlace primario entre SW-CORE-01 y SW-DIST-L1. Con RSTP, la convergencia ocurrió en menos de 5 segundos, muy por debajo del máximo de 30 segundos requerido.

```
SW-CORE-01#show spanning-tree vlan 110
VLAN0110
  Spanning tree enabled protocol rstp
  Root ID    Priority    4206
             Address     000A.F354.D2B3
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4206 (priority 4096 sys-id-ext 110)
             Address     000A.F354.D2B3
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/11                 Desg FWD 4         128.11 P2p
Gi1/0/13                 Desg FWD 4         128.13 P2p
Gi1/0/15                 Desg FWD 4         128.15 P2p
Gi1/0/19                 Desg FWD 4         128.19 P2p
Gi1/0/17                 Desg FWD 4         128.17 P2p
Fa1                     Desg FWD 3         128.29 P2p
Gi1/0/21                 Desg FWD 4         128.21 P2p
Gi1/0/23                 Desg FWD 4         128.23 P2p

SW-CORE-01#
```



```
SW-CORE-01
Physical Config CLI Attributes
IOS Command Line Interface

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/11                 Desg FWD 4         128.11 P2p
Gi1/0/13                 Desg FWD 4         128.13 P2p
Gi1/0/15                 Desg FWD 4         128.15 P2p
Gi1/0/19                 Desg FWD 4         128.19 P2p
Gi1/0/17                 Desg FWD 4         128.17 P2p
Fa1                     Desg FWD 3         128.29 P2p
Gi1/0/21                 Desg FWD 4         128.21 P2p
Gi1/0/23                 Desg FWD 4         128.23 P2p

SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW-CORE-01(config)#interface GigabitEthernet1/0/11
SW-CORE-01(config-if)#shutdown

SW-CORE-01(config-if)#end
%LINK-5-CHANGED: Interface GigabitEthernet1/0/11, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/11, changed state to down

SW-CORE-01#
%SYS-5-CONFIG_I: Configured from console by console

SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#show spanning-tree vlan 110
VLAN0110
  Spanning tree enabled protocol rstp
  Root ID    Priority    4206
             Address     000A.F354.D2B3
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    4206 (priority 4096 sys-id-ext 110)
             Address     000A.F354.D2B3
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/13                 Desg FWD 4         128.13 P2p
Gi1/0/15                 Desg FWD 4         128.15 P2p
Gi1/0/19                 Desg FWD 4         128.19 P2p
Gi1/0/17                 Desg FWD 4         128.17 P2p
Fa1                     Desg FWD 3         128.29 P2p
Gi1/0/21                 Desg FWD 4         128.21 P2p
Gi1/0/23                 Desg FWD 4         128.23 P2p

SW-CORE-01#
```

```
SW-CORE-01
Physical Config CLI Attributes
IOS Command Line Interface

SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW-CORE-01(config)#interface GigabitEthernet1/0/11
SW-CORE-01(config-if)# shutdown

SW-CORE-01(config-if)#end
%LINK-3-CHANGED: Interface GigabitEthernet1/0/11, changed state to administratively down
%LINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet1/0/11, changed state to down

SW-CORE-01#
%SYS-5-CONFIG_I: Configured from console by console

SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#show spanning-tree vlan 110
VLAN0110
  Spanning tree enabled protocol rstp
  Root ID    Priority    4206
            Address    000A.F354.D283
            This bridge is the root
            Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    4206 (priority 4096 sys-id-ext 110)
            Address    000A.F354.D283
            Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
            Aging Time 20

Interface    Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/13    Desg FWD 4      128.13 P2p
Gi1/0/15    Desg FWD 4      128.15 P2p
Gi1/0/19    Desg FWD 4      128.19 P2p
Gi1/0/17    Desg FWD 4      128.17 P2p
Fa0/1       Desg FWD 3      128.29 P2p
Gi1/0/21    Desg FWD 4      128.21 P2p
Gi1/0/23    Desg FWD 4      128.23 P2p

SW-CORE-01#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW-CORE-01(config)#interface GigabitEthernet1/0/11
SW-CORE-01(config-if)# no shutdown

SW-CORE-01(config-if)#end
%LINK-3-CHANGED: Interface GigabitEthernet1/0/11, changed state to up
%LINEPROTO-3-UPDOWN: Line protocol on Interface GigabitEthernet1/0/11, changed state to up

SW-CORE-01#
%SYS-5-CONFIG_I: Configured from console by console

SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
```

```
C:\>ping 10.10.80.22 -n 100

Pinging 10.10.80.22 with 32 bytes of data:

Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time=10ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time=35ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time=6ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Request timed out.
Reply from 10.10.80.22: bytes=32 time=12ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time=1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
```

2. Failover HSRP

Se apagó el SW-CORE-01 (router HSRP activo). El SW-CORE-02 asumió el rol activo en menos de 10 segundos. Los pings continuos mostraron pérdida mínima de 2-3 paquetes durante la transición.

```
SW-CORE-01
Physical Config CLI Attributes
IOS Command Line Interface
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#show standby brief
SW-CORE-01#
      F indicates configured to preempt.
Interface Grp Pri P State Active Standby Virtual IP
Vl10      10  110 F Active local 192.168.10.3 192.168.10.1
Vl20      20  110 F Active local 192.168.20.3 192.168.20.1
Vl21      21  110 F Active local 192.168.21.4 192.168.21.1
Vl22      22  110 F Active local 192.168.22.3 192.168.22.1
Vl23      23  110 F Active local 192.168.23.3 192.168.23.1
Vl24      24  110 F Active local 192.168.24.3 192.168.24.1
Vl25      25  110 F Active local 192.168.25.3 192.168.25.1
Vl30      30  110 F Active local 192.168.30.3 192.168.30.1
Vl31      31  110 F Active local 192.168.31.3 192.168.31.1
Vl32      32  110 F Active local 192.168.32.3 192.168.32.1
Vl33      33  110 F Active local 192.168.33.3 192.168.33.1
Vl99      99  110 F Active local 192.168.99.3 192.168.99.1
Vl110     110 110 F Active local 10.10.10.3 10.10.10.1
Vl111     111 110 F Active local 10.10.11.3 10.10.11.1
Vl120     120 110 F Active local 10.10.20.3 10.10.20.1
Vl121     121 110 F Active local 10.10.21.3 10.10.21.1
Vl122     122 110 F Active local 10.10.22.3 10.10.22.1
Vl123     123 110 F Active local 10.10.23.3 10.10.23.1
Vl130     130 110 F Active local 10.10.30.3 10.10.30.1
Vl131     131 110 F Active local 10.10.31.3 10.10.31.1
Vl132     132 110 F Active local 10.10.32.3 10.10.32.1
Vl133     133 110 F Active local 10.10.33.3 10.10.33.1
Vl140     140 110 F Active local 10.10.40.3 10.10.40.1
Vl141     141 110 F Active local 10.10.41.3 10.10.41.1
Vl142     142 110 F Active local 10.10.42.3 10.10.42.1
Vl143     143 110 F Active local 10.10.43.3 10.10.43.1
Vl150     150 110 F Active local 10.10.50.3 10.10.50.1
Vl151     151 110 F Active local 10.10.51.3 10.10.51.1
Vl152     152 110 F Active local 10.10.52.3 10.10.52.1
Vl153     153 110 F Active local 10.10.53.3 10.10.53.1
Vl160     160 110 F Active local 10.10.60.3 10.10.60.1
Vl161     161 110 F Active local 10.10.61.3 10.10.61.1
Vl162     162 110 F Active local 10.10.62.3 10.10.62.1
Vl163     163 110 F Active local 10.10.63.3 10.10.63.1
Vl170     170 110 F Active local 10.10.70.3 10.10.70.1
Vl171     171 110 F Active local 10.10.71.3 10.10.71.1
Vl172     172 110 F Active local 10.10.72.3 10.10.72.1
Vl173     173 110 F Active local 10.10.73.3 10.10.73.1
Vl180     180 110 F Active local 10.10.80.3 10.10.80.1
Vl181     181 110 F Active local 10.10.81.3 10.10.81.1
Vl182     182 110 F Active local 10.10.82.3 10.10.82.1
Vl183     183 110 F Active local 10.10.83.3 10.10.83.1
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
SW-CORE-01#
```

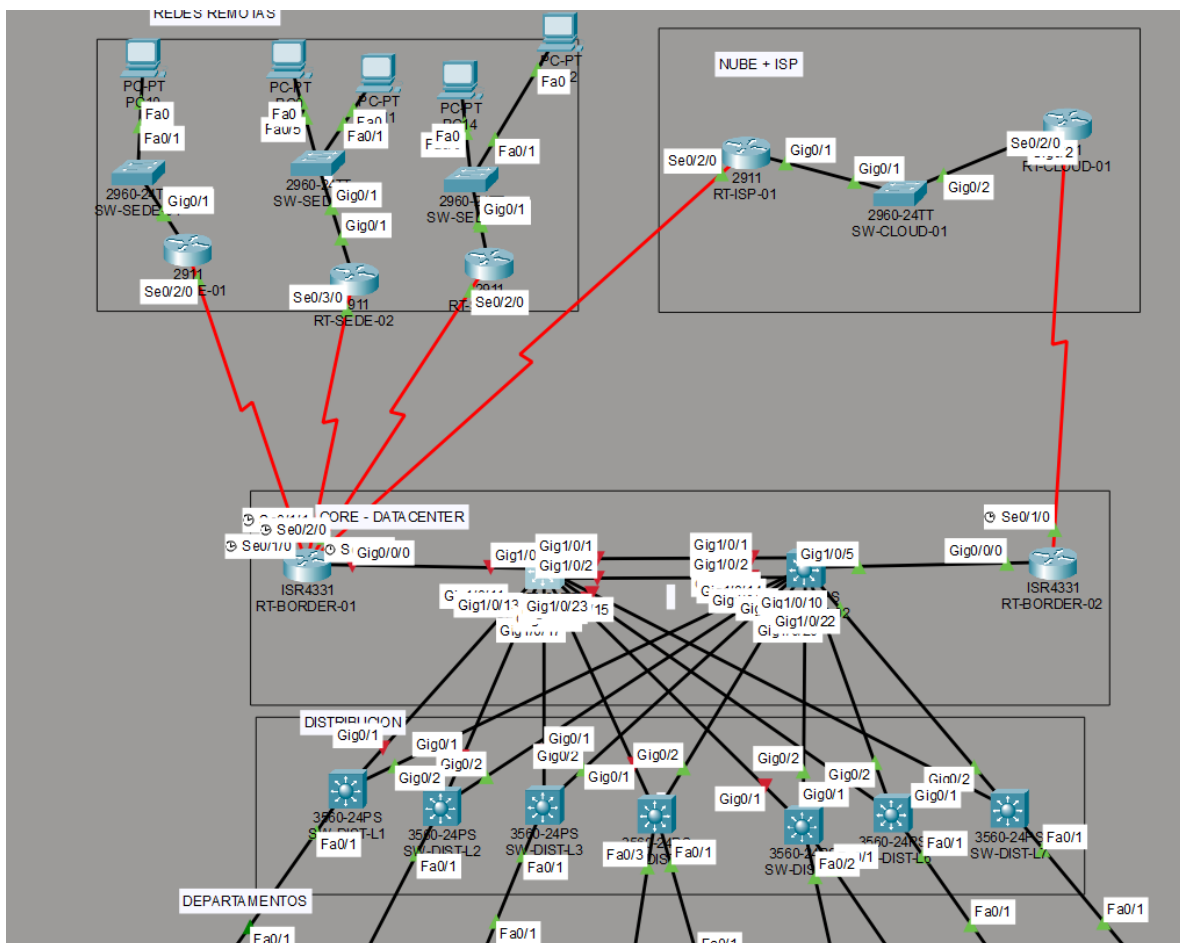
```

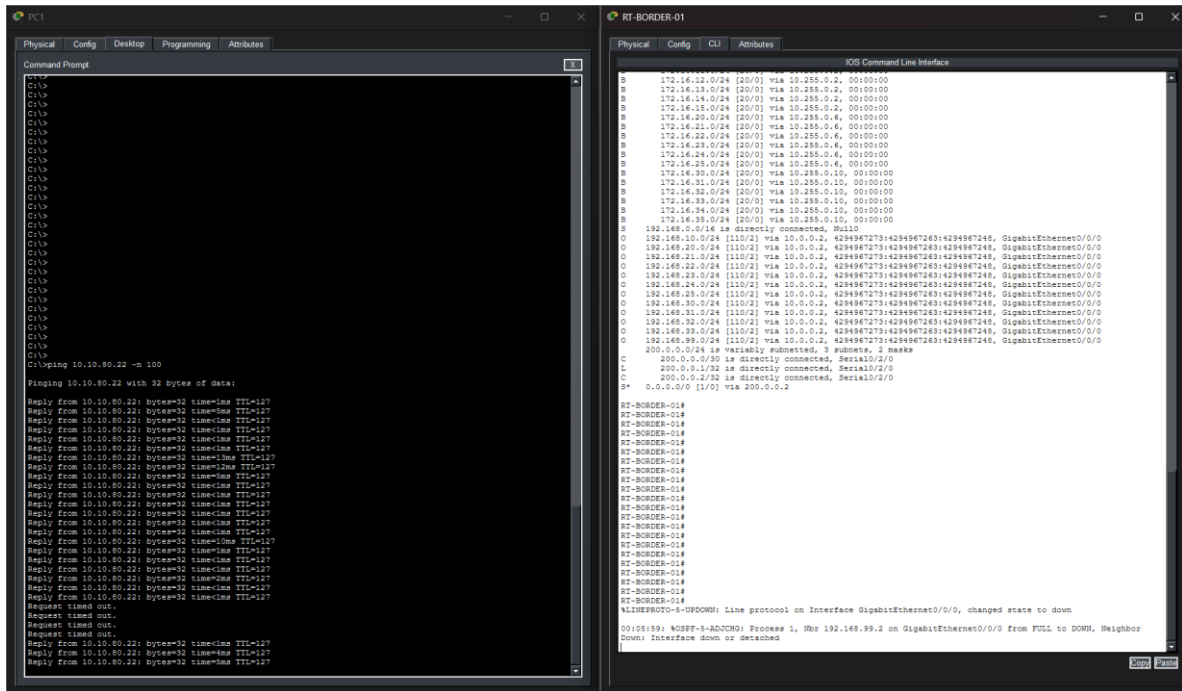
C:\>
C:\>
C:\>ping 10.10.80.22 -n 100

Pinging 10.10.80.22 with 32 bytes of data:

Reply from 10.10.80.22: bytes=32 time=1ms TTL=127
Reply from 10.10.80.22: bytes=32 time=5ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time=13ms TTL=127
Reply from 10.10.80.22: bytes=32 time=12ms TTL=127
Reply from 10.10.80.22: bytes=32 time=5ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time<1ms TTL=127
Reply from 10.10.80.22: bytes=32 time=10ms TTL=127
Reply from 10.10.80.22: bytes=32 time=1ms TTL=127

```





3. Failover WAN

Se simuló falla del enlace WAN primario apagando Serial0/2/0 en RT-BORDER-01. OSPF reconvergió y redirigió el tráfico hacia RT-BORDER-02, manteniendo conectividad con la nube.

```
RT-ISP-01
Physical Config CLI Attributes
IOS Command Line Interface
Compiled Wed 18-Jul-07 04:52 by pt_team
Image text-base: 0x2100F918, data-base: 0x24729040

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wll/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400K5
5 Gigabit Ethernet interfaces
4 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249556K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up

RT-ISP-01>
RT-ISP-01>
RT-ISP-01>
RT-ISP-01>
RT-ISP-01>en
RT-ISP-01#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 200.0.0.1 to network 0.0.0.0

C    200.0.0.0/24 is variably subnetted, 3 subnets, 2 masks
C    200.0.0.0/30 is directly connected, Serial0/2/0
C    200.0.0.1/32 is directly connected, Serial0/2/0
L    200.0.0.2/32 is directly connected, Serial0/2/0
C    200.0.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.0.1.0/30 is directly connected, GigabitEthernet0/1
L    200.0.1.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 200.0.0.1

RT-ISP-01#
```

```
RT-BORDER-01
Physical Config CLI Attributes
IOS Command Line Interface

RT-BORDER-01>en
RT-BORDER-01#conf t
Enter configuration commands, one per line. End with CNTL/Z.
RT-BORDER-01(config)#interface ser
RT-BORDER-01(config)#interface serial 0/2/0
RT-BORDER-01(config-if)#shutdown

RT-BORDER-01(config-if)#
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to administratively down

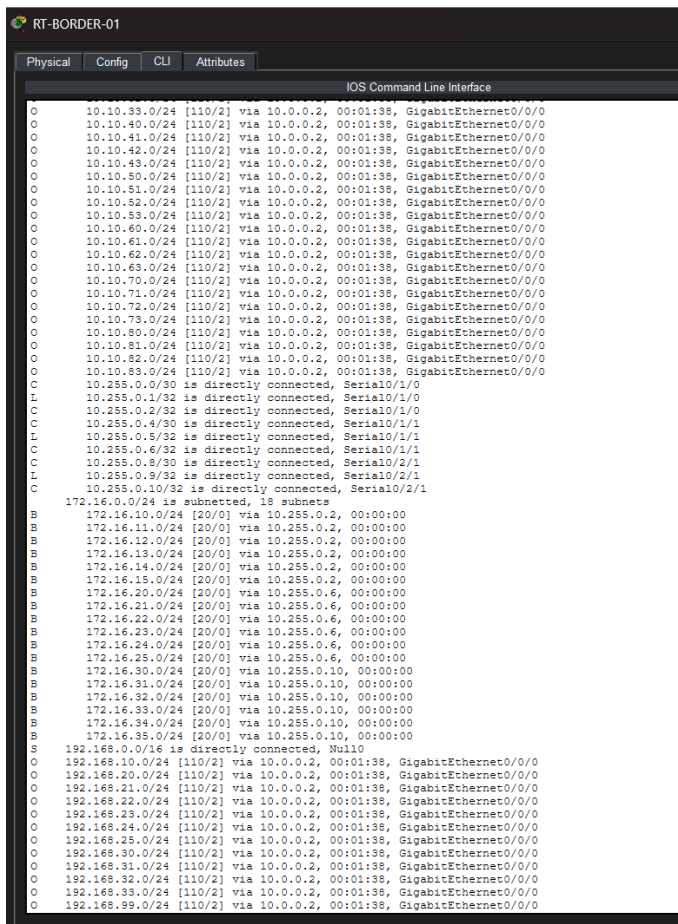
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to down
end
RT-BORDER-01#
%SYS-5-CONFIG_I: Configured from console by console

RT-BORDER-01#
RT-BORDER-01#
RT-BORDER-01#RT-BORDER-01# show ip route
^
% Invalid input detected at '^' marker.

RT-BORDER-01#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 42 subnets, 4 masks
S 10.0.0.0/8 is directly connected, Null0
C 10.0.0.0/30 is directly connected, GigabitEthernet0/0/0
L 10.0.0.1/32 is directly connected, GigabitEthernet0/0/0
O 10.10.10.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.11.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.20.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.21.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.22.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.23.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.30.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.31.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.32.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.33.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.40.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.41.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.42.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.43.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.50.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.51.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.52.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.53.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.60.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.61.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.62.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.63.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.70.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.71.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
O 10.10.72.0/24 [110/2] via 10.0.0.2, 00:00:59, GigabitEthernet0/0/0
```



Link para el video

<https://www.youtube.com/watch?v=txvtKaUOwsU>